7.1 Inverse Variation

Name _____ Period ____ Date _____

Algebra II 7.1 (30-minute plan) · Student sheet

I can…

- tell whether x and y show direct variation, inverse variation, or neither.
- write and use inverse variation equations.
- use inverse variation to solve real-life problems.

Vocabulary · fill in the missing word

Direct variation — $y = ax$, $a \neq 0$: $\frac{y}{x}$ is _____

Inverse variation — $y = \frac{a}{x}$, $a \neq 0$: the product xy is _____

Constant of variation — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Direct variation: y = _____, so y/x is _____.

Inverse variation: y = _____, so the _____ xy is constant.

Example 1

Tell whether x and y show direct variation, inverse variation, or neither.

a. $xy = 5$ b. $y = x - 4$ c. $3y = 2x$ d. $x = \frac{7}{y}$

Writing an inverse variation: To write an inverse variation equation, find a = _____, then write y = a/x.

Example 2

x and y vary inversely, and y = 6 when x = 4. Write an equation that relates x and y. Then find y when x = 3.

Part 2 · Partner practice

Partner practice

x and y vary inversely, and y = −5 when x = 3. Find y when x = −6.